

REPORT

Detailed Report for Crud Blog App.

Submitted By: Rohan Wajekar

Department: Computer Science Engineering (Artificial intelligence & Machine learning)

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BLOG CRUD APPLICATION PROJECT REPORT

1. Introduction

For this project, I developed a Blog CRUD Application to strengthen my understanding of full-stack web development and database-driven applications. The primary purpose of building this project was to understand how modern applications handle data creation, retrieval, updating, and deletion while connecting frontend interfaces with backend systems.

While developing this application, I focused on creating a platform where users can create, manage, edit, and remove blog posts efficiently. This project allowed me to gain practical experience with frontend development, backend integration, routing mechanisms, API communication, and database operations.

Rather than only learning concepts theoretically, I wanted to build something practical that demonstrates how complete applications function in real-world scenarios.

2. Objectives

The major objectives behind developing this project were:

- To build a complete full-stack blogging application
- To implement CRUD operations for managing blog content
- To understand database integration and data persistence
- To improve my understanding of backend APIs and routing systems
- To create a responsive and user-friendly application
- To gain hands-on experience in full-stack application development

3. Problem Statement

Traditional static websites require manual modification whenever content needs to be updated, which becomes inefficient for dynamic applications. Modern applications require systems where users can create, update, retrieve, and delete information easily.

To address this problem, I developed a Blog CRUD Application that allows users to manage blog content dynamically through an interactive interface. The project aims to provide a simplified content management experience while demonstrating complete data flow between frontend, backend, and database systems.

4. Technologies Used

Frontend Technologies

- HTML
- CSS
- JavaScript

Backend Technologies

- Node.js
- Express.js

Database

- MongoDB

Development Tools

- VS Code
- Git
- GitHub

5. System Architecture

The application follows a client-server architecture where user interactions pass through multiple layers before reaching the database.

User Interface



Frontend Components



Backend APIs



Database Operations



Stored Blog Information

I implemented this architecture to separate presentation logic, business logic, and database operations which makes the application easier to manage and scale.

6. Features Implemented

During development, I implemented several features including:

- Creating new blog posts
- Viewing existing blog content
- Updating previously created blogs
- Deleting blog posts
- Database storage and retrieval
- API integration between frontend and backend
- Dynamic rendering of content
- Route handling and navigation

These features helped me understand how complete CRUD applications function internally.

7. Development Methodology

Step 1

I first analyzed project requirements and planned the structure of the application.

Step 2

I designed frontend interfaces and planned user interactions.

Step 3

I developed backend routes and connected the application with the database.

Step 4

I implemented CRUD functionality and integrated APIs.

Step 5

I tested various functionalities, debugged issues, and improved application performance.

Step 6

I organized project files, documentation, and prepared the project for deployment.

8. Challenges Faced

While developing this project, I encountered several challenges.

- Managing database connectivity and data flow
- Handling backend API requests and responses
- Debugging route-related issues
- Synchronizing frontend and backend communication
- Organizing project structure efficiently

Working through these challenges improved my debugging skills and helped me better understand full-stack development workflows.

9. Results and Outcomes

The final application successfully performs all CRUD operations and allows users to manage blog content dynamically.

Through this project, I improved my understanding of:

- Full-stack application workflows
- Database operations and management
- Backend architecture and routing
- API communication
- Debugging and problem-solving techniques

This project significantly improved my practical understanding of web application development.

10. Future Improvements

Although the application currently performs its intended functionalities, I plan to improve it further by adding:

- User authentication and authorization
- User accounts and profile management
- Search and filtering functionality
- Comments section
- Blog categories and tags
- Analytics dashboard
- Cloud deployment
- Better responsiveness and UI improvements

These additions would make the application more scalable and closer to real-world blogging platforms.

11. Conclusion

Developing the Blog CRUD Application gave me valuable hands-on experience with full-stack development. Through this project, I learned how frontend interfaces communicate with backend systems and databases to create dynamic applications.

More importantly, this project helped me move beyond theoretical concepts and understand how complete applications are structured, developed, tested, and maintained.

Overall, this project served as an important step in strengthening my practical development skills and building confidence in creating larger and more scalable applications.

12.REFERENCES

[1] Blog CRUD Application Source Code Repository

GitHub Repository:

<https://github.com/rohanw2004/Blog-crud-app>

Accessed on: June 2026

Description:

This repository contains the complete source code, project structure, implementation files, frontend and backend components, and documentation related to the Blog CRUD Application project.

[2] Development Resources and Documentation

Node.js Official Documentation

<https://nodejs.org/>

Express.js Official Documentation

<https://expressjs.com/>

MongoDB Official Documentation

<https://www.mongodb.com/docs/>

MDN Web Documentation

<https://developer.mozilla.org/>

Project Developed and Maintained By:

Rohan Wajekar

GitHub Profile:

<https://github.com/rohanw2004>